CHAPTER 6

(Just Enough) Algebraic Geometry

"whatever doesn't kill you simply makes you stranger" – Joker

"I am sad to hear that one cannot present Weil's results without juggling with generic points; as a matter of fact, the unbridled abuse of generic points necessarily hides the few situations in which their use is truly essential, such as the proof that every endomorphism of the Jacobian comes from a correspondence. As Chevalley says, one feels frustrated when faced with a proof like that one.'' – Grothendieck to Serre in a letter on November 22, 1956

6.1. There are two routes to algebraic geometry. The first one goes through the theory of varieties, and the second, which is relatively modern, goes directly to the theory of schemes. In this book, we will mostly discuss the latter. However, historically speaking, varieties have been crucial for algebraic geometry too and, of course, play a significant role in many parts of mathematics (and physics!). So in the initial parts of this chapter, we will discuss the classical theory of varieties.

In order to keep this chapter a little bit self-contained, we will try to introduce a few definitions from commutative algebra as we go along.

6.2. The scheme has been standard for more than six decades. Before the language of schemes, there was the definition of algebraic variety in the pens of Weil [**24**] and Zariski (and the abstract variety in the sense of Serre). An affine variety is just a special case of a scheme. The formalization of scheme theory was done by Grothendieck (with the help of works by Serre and Chevalley) in Éléments de Géométrie Algébrique.

In this chapter, we will discuss some basics of varieties that will comfort those who have not seen them before. We will swiftly change our discussion to schemes. This will be an important chapter in Prelude to Schemes, as we will discuss most of the EGA work herein.

This is a part of Prelude of Schemes project available at `https://aayushayh.github.io/PtoS.html`.
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Textbooks [**25**, **26**] do a nice job of introducing varieties, and [**22**] is an excellent resource for algebraic geometry in the language of schemes straightaway, which will be closer to our hearts in this chapter. For the part about varieties, we will follow [**25**].

6.3 (Notation). Throughout the chapter, we fix k to be an algebraically closed field unless stated otherwise.

1. A little bit of Variety does not kill you

Definition 6.4 (Affine space over k). An affine n-space over k, denoted as A_k^n, is a set of all n-tuples of elements of k. An element P of A_k^n is called a point, and the components of $P \in A_k^n$ are called the coordinates of point P. So if $P = (a_1, \cdots, a_n)$ with $a_i \in k$, then we call a_i to be coordinates.

In a very loose sense, A_k^n is just an arbitrary geometric way to write k^n (without an origin). But we will postpone this discussion to a later stage.

6.5. Given a polynomial ring[1] over k, $A = k[x_1, \cdots, x_n]$, we can write elements of A as elements $f \in A$ which is a function $f : A_k^n \to k^n$,

$$f(P) = f(a_1, \cdots, a_n), \quad f \in A \text{ and } P \in A_k^n \tag{128}$$

and we can define the zeroes of f as a set $Z(f)$ but say we have a subset $T \subseteq A$, so define the zero set of T as the common zeroes of all the elements of T

$$Z(T) = \{P \in A_k^n | \forall f \in T, f(P) = 0\} \tag{129}$$

6.6. We will now denote the affine $n-$space A_k^n by $\mathbf{A}^n$.

Definition 6.7 (Algebraic Sets). A subset V of $\mathbf{A}^n$ is called *algebraic set* if $V = Z(T)$ for some T.

Definition 6.8. An *algebraic plane curve* over k is an algebraic set $C \subseteq \mathbf{A}^2$. We will hopefully discuss affine curves in detail.

6.9. We will define the Zariski topology on $\mathbf{A}^n$. Now, the intersection of two algebraic sets is an algebraic set, and so is the union of two algebraic sets. (See [**25**] for a proof.)

The Zariski topology on $\mathbf{A}^n$ is defined by taking the complements of algebraic sets to be open subsets. Since their intersections and unions will be open, it is a topology. And the empty set and whole space, being algebraic sets, are both open.

6.10. A non-empty subset W of a topological space X is said to be irreducible if it cannot be written as the union of two proper subsets of X and closed in Y, i.e.,

$$W = W_1 \cup W_2. \tag{130}$$

[1] A polynomial ring is a unique factorization domain (UFD).

(A curve is said to be irreducible if its polynomial equation is irreducible.)

PROPOSITION 6.11. Any closed set is a finite union of irreducible closed sets. The product of any two irreducible closed sets is irreducible again.

6.12 (EGA [16], (2.1.2)). A subspace W of a topological space is said to be irreducible if its closure $\overline{W}$ is irreducible. A subspace which is the closure of a singleton x is irreducible.

For $y \in \overline{\{x\}}$, which is same as $\overline{\{y\}} \subset \overline{\{x\}}$, that y is a *specialization* of x or that x is a *generalization* of y. If X is irreducible and there exists a point $x \in X$ such that $X = \overline{\{x\}}$, we call x a *generic point* of X. A non-empty open subset of X having x will then have x as a generic point.

A generic point is an interesting feature of algebraic geometry, and we will come to it again sometime.

DEFINITION 6.13. A *Kolmogorov* space or T_0 space is a topological space satisfying the *axiom of separation*

T_0 : For all $x, y \in X$ and $x \neq y$, there exists an open set O such that it contains one of the points x and y, but not the other (in other words, $(x \in O) \wedge (y \notin O)$ or $(x \notin O) \wedge (y \in O)$).

If Kolmogorov space admits a generic point, then there exists exactly one generic point in Kolmogorov space.

DEFINITION 6.14 (Open covering of X). An *open covering* U_α of X is a collection $U_\alpha \subset X$ of open subsets such that $\bigcup_\alpha U_\alpha = X$. A *subcover* of X will be a subcollection that still covers X. A *finite subcover* contains only finitely many sets. A *good cover*[2] is an open cover in which every nonempty finite intersection $U_i \cap U_j \cap \cdots \cap U_k$ is contractible.

DEFINITION 6.15 (Quasi-compactness). Quasi-compactness of topological space and continuous map:

(1) A topological space X is called *quasi-compact* whenever any collection of open sets of X is a finite cover of X then there exists a finite subcollection that still covers X. In other words, every open covering of X has a finite subcover.
(2) For any quasi-compact open subspace $V \subset Y$, we define a continuous map $f : X \to Y$ to be *quasi-compact* if the inverse image $f^{-1}(V)$ is quasi-compact.

6.16. We usually reserve the word 'quasi-compactness' for non-Hausdorff topological spaces (like T_0 spaces) since there are many spaces in algebraic geometry which are not Hausdorff. A quasi-compact space is compact if it is also Hausdorff.

[2]A good cover is an important ingredient in understanding the topology of finite intersections of a topological space or a manifold. More specifically, in manifolds, one adds that finite intersections are homeomorphic to an open ball.

6.17. Every nonempty subspace V of X is irreducible since the closure $\overline{V}$ is X (which is irreducible). If X admits a generic point x, then x is also a generic point of V. The proof of this can be found in (EGA [**16**], 2.1.4).

6.18 ([**16**]). Let X be an irreducible topological space, and $f : X \to Y$ a continuous map, then $\overline{f(X)}$ is irreducible. If x is a generic point of X, then $f(X)$ is a generic point of $f(X)$ and so of $\overline{f(X)}$.

6.19. Any irreducible subspace of a topological space X is contained in a maximal irreducible subspace called *irreducible component* of X which is closed.

Any irreducible space is contained in a maximal irreducible space by Zorn's lemma. Moreover, for a maximal irreducible subspace V, its closure $\overline{V}$ is also irreducible with $V \subseteq \overline{V}$, and maximality gives $V = \overline{V}$, hence V is closed.[3]

Definition 6.20 (Connectedness). A topological space X is called *connected* if X can not be written as a disjoint union

$$X = U \sqcup V \tag{131}$$

with $U, V \subseteq X$ non-empty open subsets.

Example 6.21. A non-example of connected subspace $X = \varnothing$.

Definition 6.22. A subset $V \subseteq X$ is connected if it is connected as a subspace topology. (Its closure would be connected too.)

A subspace V of X is called the *connected component* if it is contained in the maximal connected subspace.

6.23 (Lemma 5.7.2, [**11**]). Given a continuous map $f : X \to Y$ between topological spaces, if $V \subseteq X$ is a connected subset, then $f(V) \subseteq Y$ is connected as well.

Definition 6.24 (Affine Algebraic Variety). We define an *affine algebraic variety* as an irreducible closed subset of $\mathbf{A}^n$, with induced topology. Moreover, if we take an open subset of an affine variety[4], then it is called a *quasi-affine variety*.

6.25. Now, the subsets of $\mathbf{A}^n$ can be mapped to ideals in A. Given any subset $V \subseteq \mathbf{A}^n$, we define

$$I(V) = \{f \in A = k[x_1, \cdots, x_n] | f(P) = 0 \ \forall P \in V\} \tag{132}$$

Theorem 6.26 (Hilbert's Weak Nullstellansatz, from [**22**]). Let $A = k[x_1, \cdots, x_n]$ be a polynomial ring and k be an algebraically closed field (see 6.28 for why), then the maximal ideals in A are precisely those ideals of the form $(x_1 - a_1, \cdots, x_n - a_n)$, where $a_i \in k$.

[3] A closure of a subspace in a topological space is closed by definition.

[4] In French, the word 'variété' means both an algebraic variety and a differentiable manifold.

6.27. The statement of the theorem 6.26 might not be the best way to introduce Hilbert's Nullstellensatz in a progression of varieties. But a statement exists about the correspondence between ideals and varieties.

Note that in Point. 6.5, we defined the zero set of a subset of $A = k[x_1, \cdots, x_n]$. Given an ideal $J \subseteq A$, we can define a zero set, let us call it *zero locus* J, and re-notate it by

$$V(J) = \{P \in \mathbf{A}^n | \forall f \in J \ f(P) = 0\} \tag{133}$$

and when $V(J) = \varnothing$ that means that there are no points in $\mathbf{A}^n$ where polynomials in J vanish simultaneously.

6.28. There could be two situations when $V(J) = \varnothing$. Firstly, when $J = A$, then $1 \in J$ and for any point $P \in \mathbf{A}^n$, we have $1(P) \neq 0$. Therefore, $V(J) = \varnothing$ and hence no point lies in $V(J)$. One can deduce that the question of whether $V(J) =$ becomes non-trivial only if J is a proper ideal in A. Second instance when $V(J) = \varnothing$ happens when k is not algebraically closed, for example, $\mathbb{Q}, \mathbb{R}$. Given $k = \mathbb{R}$ or $k = \mathbb{Q}$ and an ideal $J = (x^2+1) \subseteq \mathbb{R}[x]$ generated by x^2+1, the zero-locus is empty $V(J) = \varnothing$. Hence, one is required to have an algebraically closed field k.

- It is also worth noting that for $J = 0$, we have $V(J) = \mathbf{A}^n$.
- If $I \subset J$, then $V(I) \supset V(J)$ which is known as inclusion-reversing.
- Since the union and intersection of algebraic sets is algebraic set (see 6.9) and any algebraic set is equal to $V(J)$ for some ideal J, we have $V(I) \cup V(J) = V(I \cap J)$ and for any collection of ideals J_α, the intersection $\bigcap_\alpha V(J_\alpha) = V(\sum_\alpha J_\alpha)$.

Now, before moving ahead, some important points.

Proposition 6.29 (See [27]). If A and B are integral domains with every element of B integral over A (in other words, B is integral over A), then A is a field iff B is a field.

Lemma 6.30 (Noether's Normalization, [2]). Let k be an algebraically closed field and $A = k[x_1, \cdots, x_n]$ a finitely generated k-algebra.[5] There exists some $m \leq n$, for which there exist elements $y_1, y_2, \cdots, y_m$ algebraically independent over k such that A is a finitely generated module over the k-subalgebra $k[y_1, y_2, \cdots, y_m]$.

Lemma 6.31 (Zariski's Lemma). Let $k \to L$ be an extension of fields such that L is of finite type over k, then L is finite over k

Theorem 6.32 (Hilbert's Weak Nullstellensatz (again), [28]). For any algebraically closed field k, for any proper ideal $J \subsetneq k[x_1, \cdots, x_n]$ we have $V(J) \neq \varnothing$.

[5]Here, we gave the statement for a polynomial ring as a finitely generated k-algebra, but Noether's normalization works for any finitely generated k-algebra.

PROOF. For the ideal J to be proper in $A = k[x_1, \cdots, x_n]$, it would be contained in a maximal ideal $\mathfrak{m} \subseteq A$ (by Zorn's Lemma) and $V(\mathfrak{m}) \subseteq V(J)$ by inclusion-reversing 6.28. So just show that $V(\mathfrak{m}) \neq \varnothing$. Let us look at the quotient $L = A/\mathfrak{m}$ and since $\mathfrak{m}$ is a maximal ideal, L is a field and L is a finitely generated k-algebra. We can take k to be a subfield of L by a natural homomorphism $k \hookrightarrow L$. Since k is an algebraically closed field, k does not have any non-trivial finite field extensions. By Lemma 6.31, we have $k \cong L$.

The quotient map (which is a projection) $\pi : A \to L = A/\mathfrak{m}$ is $x_i \mapsto a_i \in k$. The kernel of this quotient map is just $ker(\pi) = \mathfrak{m}$. Let us pick $f \in J \subseteq \mathfrak{m}$, then we have

$$f(a_1, \cdots, a_n) = f(\pi(x_1), \cdots, \pi(x_n)) = \pi(f) = 0 \tag{134}$$

and hence $P = \{a_1, \cdots, a_n\} \in \mathbf{A}^n$ lies in $V(J)$.

REMARK 6.33. The statements in 6.26 and 6.32 are actually the same. If it is not obvious, we will see it later.

PROPOSITION 6.34. An algebraic set is irreducible if and only if its ideal (as defined in 6.25) is a prime ideal.

PROOF. To show this, using above discussions, we also know that there exists a one-to-one inclusion-reversing correspondence between algebraic sets and ideals (via Nullstellensatz) by $Y \mapsto I(Y)$ and $\mathfrak{a} \mapsto Z(\mathfrak{a})$ where $I(Y)$ is the ideal of an algebraic set and $Z(\mathfrak{a})$ is the algebraic set of a (radical) ideal $\mathfrak{a}$.

To prove this (from one side), one has to show that $I(V)$ is a prime ideal. We follow [**25**]. Let $fg \in I(V)$, then using the inclusion-reversing property, $V \subseteq Z(fg) = Z(f) \cup Z(g)$. We can write it as $V = (V \cap Z(f)) \cup (Y \cap Z(g))$ but since V is irreducible algebraic set, we have either $V = (V \cap Z(f))$ which means $V \subseteq Z(f)$ or $V = (V \cap Z(g))$ which means $V \subseteq Z(g)$, which means that either $f \in I(V)$ or $g \in I(V)$ and hence, $I(V)$ is a prime ideal.

Now, the other side. Let us say that $\mathfrak{p}$ is a prime ideal and say $Z(\mathfrak{p}) = V_1 \cup V_2$, then $\mathfrak{p} = I(V_1) \cap I(V_2)$, so we have either $\mathfrak{p} = I(V_1)$ or $\mathfrak{p} = I(V_2)$ and hence, we have either $Z(\mathfrak{p}) = V_1$ or $Z(\mathfrak{p}) = V_2$ and so $Z(\mathfrak{p})$ is irreducible.

DEFINITION 6.35. For an algebraic set $V \subset \mathbf{A}^n$, we define the affine coordinate ring $A(V) = A/I(V)$.

REMARK 6.36. This affine coordinate ring is an integral domain if and only if V is irreducible.

Now, we can move to some important topology of the varieties.

DEFINITION 6.37 (Noetherian Topological Space). For a topological space X, if it satisfies a descending chain condition for closed subsets of X

$$V_1 \supseteq V_2 \supseteq V_3 \cdots \tag{135}$$

such that there exists an integer r with $V_r = V_{r+1} = \cdots$ then such a topological space is called *Noetherian.*

6.38. There is a hereditary property. A subspace of a Noetherian space is also Noetherian [**1**]. Moreover, any Noetherian space is quasi-compact (see Def. 6.15), and any space where all open balls are quasi-compact is Noetherian [**1**, **16**].

Example 6.39. The affine space $\mathbf{A}^n$ is Noetherian. Since, $A = k[x_1, \cdots, x_n]$ is a Noetherian ring, the chain of ideals in A saturates and from Nullstellensatz, we have $V_i = Z(I(V_i))$ and $I(V_i)$ is stationary in A then chain of V_i will also be stationery.

Definition 6.40. A topological space X is called *locally Noetherian* if every point has a neighborhood which is Noetherian.

Remark 6.41 (Stacks [**11**], Lemma 04MF). If X is a locally Noetherian topological space, then X is locally connected.

Lemma 6.42 (Stacks [**11**], Lemma 0052). Let X be a Noetherian topological space, then

- Any subset of X with induced topology is Noetherian.
- Topological space X has finitely many irreducible components.
- Each irreducible component of X contains a nonempty open of X.

Lemma 6.43 (Stacks [**11**], Lemma 04Z8). Let $f : X \to Y$ be a continuous map of topological spaces, then

- If X is Noetherian, then so is $f(X)$.
- If X is locally Noetherian and f is open map, then $f(X)$ is locally Noetherian.

Proposition 6.44. Now, every algebraic set in $\mathbf{A}^n$ can be written uniquely as a union of varieties, where no two contain each other.

The discussions above make it clear.

Definition 6.45 (Krull dimension of X). Let X be a topological space. A chain of irreducible closed subsets of X is a sequence $Z_0 \subset Z_1 \subset \cdots \subset Z_n$ with each Z_i irreducible and $Z_i \neq Z_{i+1}$. Then n is the *length* of the chain. We define the Krull dimension of X as

$$dim(X) = \text{suplengths of chains of irreducible closed subsets} \tag{136}$$

Dimension can be infinite. One can define the dimension of the empty set *varnothing* as $dim(\varnothing) = -\infty$, as there are no chains of irreducible closed subsets at all for $\varnothing$.

6.46. If $V \subseteq X$, then $\dim V \leq \dim X$.

Definition 6.47 (Height of prime ideals). Let R be a ring, then for chain of length n of (distinct) prime ideals $\mathfrak{p}_0 \subset \mathfrak{p}_1 \subset \cdots \subset \mathfrak{p}_n = \mathfrak{p}$ such that $\mathfrak{p}_i \neq \mathfrak{p}_{i+1}$ for $i = 0, \cdots, n-1$, we define the *height* of $\mathfrak{p}$ to be the supremum of all the integers n. Sometimes, it is also called the *codimension* of a $\mathfrak{p}$.

DEFINITION 6.48 (Krull dimension of a ring). We define the *Krull dimension* of a ring A to be the supremum of the heights (see Def. 6.47) of all prime ideals.

DEFINITION 6.49. Given a prime ideal $\mathfrak{p}$ in R, the dimension of the local ring $R_{\mathfrak{p}}$ is the height of $\mathfrak{p}$.

PROPOSITION 6.50. If $V \subseteq \mathbf{A}^n$ is an affine algebraic set, then the dimension of V is the dimension of its affine coordinate ring $A(V)$ (see Def. 6.35)

$$\dim V = \dim A(V) \tag{137}$$

PROOF. Krull dimension of V is defined using Def. 6.51, so it is the supremum of lengths of chains of irreducible closed subsets

$$Z_0 \subset Z_1 \subset \cdots Z_n \subset V \tag{138}$$

which under Nullstellensatz follows a reversing-inclusion bijection with the prime ideals of the coordinate ring $A(V)$ (see 6.34), so $Z_i \mapsto I(Z_i)$, which are prime ideals and reverse the inclusion, so

$$\mathfrak{p}_n \subset \mathfrak{p}_{n-1} \subset \cdots \subset \mathfrak{p}_1 \subset \mathfrak{p}_0 \tag{139}$$

is a chain of prime ideals in ring $A = k[x_1, \cdots, x_n]$. All of the primes contain $I(V)$, hence a chain

$$\mathfrak{p}_n/I(V) \subset \mathfrak{p}_{n-1}/I(V) \subset \cdots \subset \mathfrak{p}_1/I(V) \subset \mathfrak{p}_0/I(V) \tag{140}$$

which is a chain of prime ideals in $A(V)$, which is the affine coordinate ring.

Thus, the chains of irreducible closed subsets of V and chains of prime ideals in its affine coordinate ring $A(V)$ are in bijection, so the lengths of corresponding chains are equal. Therefore, the supremum in both definitions of 6.45 and 6.48 are equal, hence *up to reversal of order*

$$\dim V = \dim A(V) \tag{141}$$

Conversely, for a chain of prime ideals in $A(V)$

$$\mathfrak{p}_0 \subset \mathfrak{p}_1 \subset \cdots \subset \mathfrak{p}_n \tag{142}$$

When taking inverse images in A and then its zero loci in V, one gets a chain of irreducible closed subsets of V with the same length.

DEFINITION 6.51 (As [**25**]). We define the *dimension* of (quasi)-affine variety as its dimension as a topological space.

EXAMPLE 6.52. The dimension of $\mathbf{A}^1$ is 1 as the maximal length chains are like $P \subset \mathbf{A}^1$. In fact, the dimension of $\mathbf{A}^n$ is n as there exist a chain of length n (can be seen using 6.50)

$$V(x_1, \cdots, x_n) \subset V(x_1, \cdots, x_{n-1}) \cdots \subset V(x_1) \subset \mathbf{A}^n \tag{143}$$

6.53. Now, a couple of recipes before 6.61 about dimensions.

PROPOSITION 6.54. Let A be a finitely generated integral k-algebra (for k an arbitrary field). Then there exist algebraically independent elements $y_1, y_2, \cdots, y_n \in A$ such that A is integral over[6] $k[y_1, \cdots, y_n]$.

Also see Lemma. 6.30.

COROLLARY 6.55. Since A is integral over $k[y_1, \cdots, y_n]$, we have

$$\dim A = \dim k[y_1, \cdots, y_n] = n \tag{144}$$

as dimension is invariant under integral extensions. The dimension of A and $\dim k[y_1, \cdots, y_n]$ are just the Krull dimension (also see Example. 6.52 as dimension of $\mathbf{A}^n$ is n).

DEFINITION 6.56. For an integral domain[7] A, we define the *quotient field* or *field of fractions* or *fraction field* of A as $K(A)$

$$K(A) = \frac{a}{b} \mid a \in A, b \in A \backslash 0 \ / \sim \tag{145}$$

where the equivalence relation identifies $a/b = a'/b'$ if and only if $ab' = ba'$.

6.57. There is an embedding $A \to K(A)$. A common example is that the field of fractions for the ring of integers $\mathbb{Z}$ is $\mathbb{Q}$.

There is a canonical isomorphism between a field and its field of fractions.

Moreover, the quotient field or field of fractions $K(A)$ has a universal property. Given an injective ring homomorphism $\phi : A \to F$ where F is a field, then there is a unique map $\phi : K(A) \to F$ such that the diagram commutes

$[\ K(A) \rightarrowtail \ ]$

DEFINITION 6.58. Recall that a field extension L/K is called a *transcendental extension* if there exists an element in L which is transcendental over K. In other words, L/K is not an algebraic extension.[8]

We define *transcendence basis* of L/K to be the maximal subset of algebraically independent elements $S = \{s_1, s_2, \cdots, s_n\} \subseteq L$. The cardinality or the size of S is called *transcendence degree* of L/K denoted as $\text{tr.deg}_K L$, see Def. 6.58.

EXAMPLE 6.59. Trivial examples of transcendental extensions include $\mathbb{C}/\mathbb{Q}$ and $\mathbb{R}/\mathbb{Q}$.

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[6]We say that A is integral over $k[x_1, \cdots, x_n]$ if for all $y \in A$ there is a monic polynomial $f(t) \in k = [x_1, \cdots, x_n][t]$ such that $f(y) = 0$.

[7]Without the notion of integral domain, the multiplication on $K(A)$ defined by $\frac{a}{b} \cdot \frac{c}{d} = \frac{ac}{bd}$ where $bd \neq 0$ only when A is integral domain.

[8]An algebraic extension L/K means that every element in L is a root to a non-zero polynomial with coefficients in K. For example, $\mathbb{C}/\mathbb{R}$ is an algebraic extension.

Moreover, a field extension L/K is algebraic only when its transcendence basis if $\varnothing$.

LEMMA 6.60. Let A be integral over $k[y_1, \cdots, y_n]$ as 6.54 so $k[y_1, \cdots, y_n] \subseteq A$, then their quotients fields would be $K(k[y_1, \cdots, y_n]) \subseteq K(A)$. We know that $K(k[y_1, \cdots, y_n]) = k(y_1, \cdots, y_n)$, which is the field of rational functions, which is a transcendental extension of the base field k with $\text{tr.deg}_k k(y_1, \cdots, y_n) = n$. Moreover, $K(A)$ is algebraic over $k(y_1, \cdots, y_n)$, hence $\text{tr.deg}_k K(A) = n$ (Theorem 1.8A in [**25**]). (One may see that as when passing to the quotient field, we write algebraic over instead of integral over.)

PROOF. The proof of the last statement is as follows. Given a tower of fields $L/K/k$, we have

$$\text{tr.deg}_k L = \text{tr.deg}_K L + \text{tr.deg}_k K. \tag{146}$$

Now for $K(A)/k(y_1, \cdots, y_n)/k$, we similarly write

$$\text{tr.deg}k K(A) = \text{tr.deg}k(y_1, \cdots, y_n) K(A) + \text{tr.deg}k k(y_1, \cdots, y_n). \tag{147}$$

and since $K(A)$ is algebraic over $k(y_1, \cdots, y_n)$, we have $\text{tr.deg}k(y_1, \cdots, y_n)K(A) = 0$ and

$$\text{tr.deg}_k K(A) = \text{tr.deg}_k k(y_1, \cdots, y_n) = n \tag{148}$$

THEOREM 6.61. Let k be a field, and let A be an integral domain which is finitely k-generated algebra, then

$$\dim A = \text{tr.deg}_k K(A) \tag{149}$$

PROOF. See discussions 6.30, 6.54—6.60.

6.62. A nice slogan to take away is that *Krull dimension is invariant under integral extensions and the transcendence degree is invariant under algebraic extensions.*

THEOREM 6.63. Let k be a field, and let A be an integral domain which is finitely k-generated algebra and $\mathfrak{p}$ be a prime ideal in A, then

$$\text{height } \mathfrak{p} + \dim A/\mathfrak{p} = \dim A \tag{150}$$

PROOF. Proof uses the Noether normalization and Theorem. 6.61 and can be found in [**29**]. Note that $A/\mathfrak{p}$ is also an integral domain which is a finitely k-generated algebra, so $\dim A/\mathfrak{p} = \text{tr.deg}_k K(A/\mathfrak{p})$.

6.64. Any nonempty open subset of an irreducible space is irreducible and dense.

As a consequence of Theorem. 6.63 (see [**25**]), for a quasi-affine variety V, we have $\dim V = \dim \overline{V}$.

PROPOSITION 6.65. A variety $V \subseteq \mathbf{A}^n$ has dimension $n-1$ if and only if it is the zero set of a single nonconstant irreducible polynomial $Z(f)$ in $A = k[x_1, \cdots, x_n]$.

PROOF. We only give one side (see [**25**]). We know that $Z(f)$ is a variety for an irreducible $f \in A$. Moreover, (from Krull's Principal Ideal Theorem), the height of $\mathfrak{p}$ containing f is 1. From Theorem. 6.63, $\dim V = n - 1$.

Projective Varieties.

DEFINITION 6.66 (Projective Space). A projective space $\mathbf{P}^n$ (or $\mathbf{P}^n_k$) over algebraically closed field k is defined to be set of equivalence classes of $n+1$-tuples $(a_0, \cdots, a_n)$ where $a_i \in k$ under the equivalence relation

$$(a_0, \cdots, a_n) \sim (\lambda a_0, \cdots, \lambda a_n) \tag{151}$$

in other words, two points on a line in $\mathbf{P}^n$ are defined to be the same.

It can also be defined as a quotient space

$$\mathbf{P}^n = \mathbf{A}^{n+1} \backslash \{0, \cdots, 0\} \tag{152}$$

DEFINITION 6.67. An element P of $\mathbf{P}^n$ is called a *point* and the $(n+1)$-tuple in the equivalence class is called a *set of homogeneous coordinates of* P.

6.68. We are to use homogeneous polynomials, and unlike affine varieties, we will need graded rings for projective varieties. Let us recall the definition.

DEFINITION 6.69 (Graded Ring). A ring S is defined to be a *graded ring* if there is a decomposition of its underlying additive group into

$$S = \bigoplus_{d \geq 0} S_d \tag{153}$$

where S_d are abelian groups of homogeneous element of degree d such that for any $d, e \geq 0$, we have $S_d S_e \subseteq S_{d+e}$.

The direct sum decomposition is often called *grading*.

DEFINITION 6.70. An ideal $\mathfrak{a}$ in a graded ring $S = \bigoplus_{d \geq 0} S_d$ is called *homogeneous ideal* such that

$$S = \bigoplus_{d \geq 0} (\mathfrak{a} \cap S_d) \tag{154}$$

This means that $\mathfrak{a}$ is homogeneous ideal if it is homogeneous if and only if it can be generated by a set of homogeneous elements.[9]

6.71. In order to define the projective varieties, we have to define the vanishing set of polynomials $f \in k[x_1, \cdots, x_n]$, but this does not make sense for projective space unless we make the polynomial ring into a graded ring. We have to make sure that we have homogeneous polynomials (homogeneous ideals) since the points in projective space have an equivalence relation (see Def. 6.66).

We could define a function $f : \mathbf{P}^n \to \{0, 1\}$ such that $f(P) = 0$ if $f(a_0, \cdots, a_n) = 0$ and $f(P) = 1$ if $f(a_0, \cdots, a_n) \neq 0$.

[9] As usual, a homogeneous ideal $\mathfrak{a}$ is prime if for any two homogeneous elements $f, g \in \mathfrak{a}$ then $f \in \mathfrak{a}$ or $g \in \mathfrak{a}$.

Then, for any set T of homogeneous elements of S, the vanishing set is

$$Z(T) = \{P \in \mathbf{P}^n \mid \forall f \in T, f(P) = 0\} \tag{155}$$

and if subset $V \subset \mathbf{P}^n$ is $V = Z(T)$, then V is called an *algebraic set*.

Moreover, taking the open sets (as complements of closed algebraic sets V), we define the Zariski topology on $\mathbf{P}^n$. The discussion about reproducibility and dimensions follows from the affine space.

DEFINITION 6.72. We define a *projective algebraic variety* as an irreducible closed subset of $\mathbf{P}^n$. An open subset of a projective variety is called *quasi-projective variety.*

DEFINITION 6.73. A smooth projective curve of genus one is called *elliptic curve.*

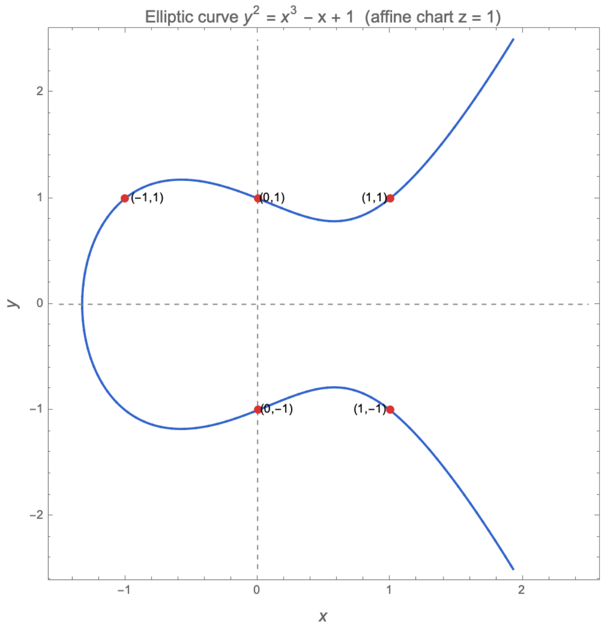

FIGURE 1. An elliptic curve $y^2 = x^3 - x + z$ with $x = 1$ affine chart.

6.74. For any subset $V \subseteq \mathbf{P}^n$, the homogeneous ideal $I(V)$ in S is

$$I(V) = f \in S \mid \forall P \in V,\ f(P) = 0 \text{ and f is homogeneous} \tag{156}$$

Like earlier in affine varieties, we define the coordinate ring of V to be $S(V) = S/I(V)$.

EXAMPLE 6.75. A projective hypersurface V is an algebraic variety in $\mathbf{P}^n$ with co-dimension 1 (which means it has $n-1$ dimension).

EXAMPLE 6.76. The zero set H_f of a linear homogeneous polynomial $f \in S$ is called *hyperplane.*

DEFINITION 6.77. A closed subvariety of $\mathbf{P}^n$ defined by an ideal generated by degree one homogeneous polynomials is called a *linear* subspace.

6.78. One claims that every (quasi)projective variety has an open covering by (quasi)affine varieties. (See Def. 6.14 for open covering.)

We need to see how every projective space admits an open covering by affine spaces.

For every homogeneous coordinate x_i in $\mathbf{P}^n$, the closed zero set is H_i for $i = 0, \cdots, n$, hence, there would be $(n+1)$ many U_i open sets in $\mathbf{P}^n$

$$U_i = \{[x_0 : \cdots : x_n] \in \mathbf{P}^n | x_i \neq 0\}. \tag{157}$$

For any non-zero homogeneous point, at least one coordinate in $\mathbf{P}^n$ is non-zero so it is in U_i and $\bigcup_{i=0}^n U_i = \mathbf{P}^n$. We define a map ϕ_i

$$\phi_i : U_i \to \mathbf{A}^n \tag{158}$$

$$[x_0 : \cdots : x_n] \mapsto \left\{ \frac{x_0}{x_i}, \cdots, \frac{x_{i-1}}{x_i}, 1, \frac{x_{i+1}}{x_i}, \cdots, \frac{x_n}{x_i} \right\} \tag{159}$$

where we omit 1 in affine coordinates. In fact, ϕ_i is a bijection (by just inserting 1 at i^{th} location)

$$x_0, \cdots, x_{i-1}, x_{i+1} \cdots, x_n \mapsto [x_0 : \cdots : x_{i-1} : 1 : x_{i+1} : \cdots : x_n] \tag{160}$$

and hence $U_i \cong \mathbf{A}^n$.

PROPOSITION 6.79. Every projective space $\mathbf{P}^n$ admits an open covering by affine spaces $\mathbf{A}^n$.

PROOF. See above.

PROPOSITION 6.80. The map $\phi_i : U_i \to \mathbf{A}^n$ is a homeomorphism between the induced topology of U_i and the Zariski topology of $\mathbf{A}^n$.

PROOF. We just have to show that ϕ and ϕ^{-1} map (as defined above) closed sets to closed sets. You may see Proposition 2.2 in [**25**] for a proof.

COROLLARY 6.81. Every projective variety is locally affine and is covered by affine charts.

DEFINITION 6.82. We define $PGL_{n+1}(k)$ to be the group

$$PGL_{n+1}(k) = GL_{{n+1}(k)/k^*} (161)$$

which is quotienting $GL_{n+1}(k)$ by the central scalars. Two invertible matrices are identified when they differ by a nonzero scalar. The group $PGL_{n+1}(k)$ is called *Projective General Linear Group*.

PROPOSITION 6.83 (See exercise in Vakil). The group $PGL_{n+1}(k)$ is the automorphism group of the projective space $\mathbf{P}^n$.

Now, we wish to define the category of varieties. If you have not seen the definition of category, then see [**11**, Definition 0014].

DEFINITION 6.84 (Regular Function). Let $Y \subseteq \mathbf{A}^n$ be an affine variety. A function $f : Y \to k$ (where k is the ground field) is called a *regular function* at point $p \in Y$ if there exists an open neighborhood $U \subseteq Y$ and there exists polynomials $g, h \in k[x_1, \ldots, x_n]$ such that $f = g/h$ is defined on U and h is nowhere zero on U.

A function is regular if it is regular at every point of Y.

EXAMPLE 6.85. A function which takes $x \mapsto 1/x$ on $\mathbf{A}^1_k \backslash \{0\} \subseteq \mathbf{A}^1_k$ is a regular function.

PROPOSITION 6.86. A regular function $f : V \to \mathbf{A}^1_k$ is continuous where $\mathbf{A}^1_k$ is taken in its Zariski topology.

DEFINITION 6.87. Let $Y \subseteq \mathbf{P}^n$ be a projective variety. A function $f : Y \to k$ is called a *regular function* at point $p \in Y$ if there exists an open neighborhood $U \subseteq Y$ and there exists homogeneous polynomials of same degree $g, h \in k[x_0, \ldots, x_n]$ such that $f = g/h$ is defined on U and h is nowhere zero on U.

A function is regular if it is regular at every point of Y.

DEFINITION 6.88 (Rational Function). A *rational function* $f : X \to Y \subseteq \mathbf{A}^n$ is an n-tuple of rational functions $f_1, \ldots, f_n \in K(X)$ such that all f_i are regular at all $x \in X$, we say that f is regular at x, $(f_1(x), \ldots, f_n(x)) \in Y$. The image under f is

$$f(X) = \{f(x) \mid x \in X\} \tag{162}$$

such that f is regular at x. Where $K(X)$ is the field of rational functions, see Def. 6.89.

DEFINITION 6.89. For an irreducible variety X, we define $K(X)$ to be the *function field* where the elements are equivalence classes of pairs (U, f) where $U \subseteq X$ and non-empty and f is a regular function on U. Since U and $V \subseteq X$ are non-empty, the intersection $U \cap V$ is non-empty. We identity two pairs (U, f) and (V, g) if $f = g$ on $U \cap V$.

REMARK 6.90. For an affine variety $X = \operatorname{Spec} A$, the function field $K(X)$ (in Def. 6.89) and field of fractions $k(A)$ are canonically isomorphic (in Def. 6.56).

PROPOSITION 6.91 (Locality). Let f be a regular function on Y, $f : Y \to k$, and let U_i be an open cover of Y. If f is regular at each restriction U_i, then $f|_{U_i}$ is regular on Y.

DEFINITION 6.92 (Category of Varieties, Hartshorne [**25**]). As almost always, let k be an algebraically closed field. We define a category of varieties as the following data. The objects of the categories are affine varieties, quasi-affine varieties, projective varieties, or quasi-projective varieties. We define a morphism $\phi : X \to Y$ between varieties X, Y as a continuous map such that for $V \subseteq Y$ and for a regular function $f : V \to k$, the pullback $f \circ \phi : \phi^{-1}(V) \to k$ is regular.

The composition of morphisms is a morphism. Moreover, identity morphisms exist as well. For morphisms $\phi : X \to Y$ and $\psi : Y \to X$, we have $\phi \circ \psi = \mathbb{1}_Y$ and $\psi \circ \phi = \mathbb{1}_X$. The category of varieties is usually denoted by $\underset{\sim}{\mathbf{Var}_k}$.

Definition 6.93 (Birational Map). A morphism between irreducible varieties $\phi : X \to Y$ is called *birational map* if there exists an inverse rational map $\psi : Y \to X$ such that both $\psi \circ \phi = \mathbb{1}_X$ and $\phi \circ \psi = \mathbb{1}_Y$ are defined (on open subset). In this case, we say X, Y varieties are *birational* or *birationally isomorphic*.

6.94. For X, Y irreducible varieties, note that $\phi(X)$ is dense in Y and $\psi(Y)$ is dense in X. If ϕ and ψ are inverse rational maps, then the image contains a non-empty subset of the co-domain, and it is dense.

Proposition 6.95. Let X, Y be irreducible varieties, then the following statements are equivalent

(1) X and Y are birationally isomorphic.
(2) There exist open non-empty subsets $U \subset X$ and $Y \subset V$ which are isomorphic.
(3) The inclusion of fields $K(X) \to K(Y)$ is an isomorphism. In other words, function fields $K(X)$ and $K(Y)$ are isomorphic as k-algebras.

Proof. It follows from the definition that (1) $\Leftrightarrow$ (2). To prove (1) $\Leftrightarrow$ (3), we define that any rational map $\phi : X \to Y$ where $\phi(X)$ is dense in Y, then the map is called *dominant* and the inclusion of function fields is an isomorphism of k-algebra (to see the proof, see these notes by Vakil[10]).

Remark 6.96. Birational equivalences between varieties are used in the classification problem, for instance, see [**30**], and this field is called *birational geometry*. So we can classify algebraic curves up to birational equivalence.

More on birational geometry will be added in future.

Definition 6.97 (Affine Hypersurface). Let k be a ground field. An *affine hypersurface* is the zero-set of a single non-constant polynomial $F \in k[x_1, x_2, \cdots, x_n]$

$$V(F) = \{x \in \mathbf{A}^n | F(x) = 0\} \tag{163}$$

Remark 6.98. Let $Y = V(F) \in \mathbf{A}^n$ be an affine hypersurface. The quotient ring

$$\frac{k[x_1, x_2, \cdots, x_n]}{I(F)} \tag{164}$$

[10] https://math.stanford.edu/~vakil/725/class13.pdf

is an integral domain if and only if $I(F)$ is a prime ideal which means F must be an irreducible polynomial (see 6.34).

THEOREM 6.99. Any irreducible variety is birational to an affine hypersurface.

PROOF. Let X be an irreducible variety. One can prove it by showing that the function field $K(X)$ is isomorphic to the function field of hypersurface (see. 6.95). Since X is of finite type over k, the function field $K(X)$ is a finitely generated as a field extension of k. Let $d = dimX = \text{tr.deg}_k K(X)$. We can choose a transcendence basis $t_1, t_2, \cdots, t_d$ of $K(X)$ over k, hence $K(X)$ is algebraic over $k(t_1, t_2, \cdots, t_d)$. One may choose additional generators for $K(X)$ as it is finitely generated as a field extension of k. But the key step is to reduce to one additional generator (by replacing two of them by a single k-linear combination). Using induction, we get

$$K(X) = k(t_1, t_2, \cdots, t_d, y_{d+1}) \tag{165}$$

for some y_{d+1} algebraic over $k(t_1, t_2, \cdots, t_d)$. This is the primitive element argument.

At this point, y_{d+1} satisfies an irreducible polynomial $F(T) \in k(t_1, t_2, \cdots, t_d)[T]$. After clearing denominators, we obtain a polynomial $H(T_1, T_2, \cdots, T_d, T) \in k[T_1, T_2, \cdots, T_d, T]$ such that $H(t_1, t_2, \cdots, t_d, y_{d+1}) = 0$. We define an affine hypersurface $Y = V(H) \subset \mathbf{A}^{d+1}$. The coordinate ring $A(Y) = \frac{k[T_1, T_2, \cdots, T_{d+1}]}{I(H)}$ is an integral domain since H is irreducible (and $I(V)$ is prime ideal). Moreover its field of fraction is the function field $K(Y)$ where there exists an isomorphism that $K(X) \cong K(Y)$. Hence X and Y are birational.

This proof is adopted from the notes available at https://sites.math.rutgers.edu/courses/535/535-f16/hyper.pdf.

DEFINITION 6.100 (Ring of regular functions, [25]). For a variety X, we denote the ring of all regular functions on Y as $\mathcal{O}(X)$. This ring is a $k-$algebra. For a point P on X, we define the *local ring* $\mathcal{O}_P(X)$ as the ring of germs of regular functions near the point P. Concretely, an element of $\mathcal{O}_P(X)$ is represented by a pair (U,f) such that U is an open neighborhood of P and f is a regular function on U and we identify two pairs (U, f) and (V, g) if $f = g$ on $U \cap V$.

REMARK 6.101. If X is affine, then $\mathcal{O}(X) \cong A(X)$ where $A(X)$ is the coordinate ring. For affine X, the local ring $\mathcal{O}_P(X)$ is the localization of the coordinate ring at the maximal ideal $\mathfrak{m}_P$, hence $\mathcal{O}_P(X) \cong A(Y)_{\mathfrak{m}_P}$.

We will later redefine these definitions of $\mathcal{O}(X)$ as global sections of the structure sheaf, while $\mathcal{O}_P(X)$ as the stalk at P.

6.102. Given a variety X and a point P on it, we have define three rings: $\mathcal{O}(X), \mathcal{O}_P(X), K(X)$. Under isomorphism of varieties $X \cong Y$ (and $P \mapsto Q$), we have

$$\mathcal{O}(X) \cong \mathcal{O}(Y) \tag{166}$$

$$\mathcal{O}_P(X) \cong \mathcal{O}_Q(X) \tag{167}$$

$$K(X) \cong K(Y) \tag{168}$$

which is because an isomorphism $X \to Y$ pullback maps on regular functions (and pullback at the level of local rings). Hence, these three rings are isomorphism invariant of the variety X and the point P.

Remark 6.103. If X is projective ($X \subseteq \mathbf{P}^n$), then $\mathcal{O}(X) \cong k$. See Theorem. 3.4 in [**25**].

Proposition 6.104. For the open vanishing set $U_i \subseteq \mathbf{P}^n$, we have an isomorphism of varieties $\phi_i : U_i \to \mathbf{A}^n$.

6.105. We have already discussed how projective space $\mathbf{P}^n$ admits an open covering by affine spaces $\mathbf{A}^n$ (see Proposition. 6.79). We have seen the homeomorphism between each $U_i \subseteq \mathbf{P}^n$ and $\mathbf{A}^n$.

The idea in Proposition. 6.104 goes beyond the change of coordinate system. One can realize that a projective space can be assembled by gluing together affine spaces along overlapping regions. Such ideas, which has its genesis in Grothendieck's work, will be frequently studied in Prelude to Schemes. In fact, as we will see about the definition of schemes.

Theorem 6.106. Let X be an arbitrary variety (affine, (quasi)projective) and $Y \subseteq \mathbf{A}^n$ be an affine variety, then there is a natural bijection

$$\alpha : Hom_{\text{Var}}(X, Y) \xrightarrow{\sim} Hom_{\text{k-alg}}(A(Y), \mathcal{O}(X)) \tag{169}$$

where $\mathcal{O}(X)$ is the ring of global regular functions on X and $A(Y) = k[x_0, \cdots, x_n]/I(Y)$ is the coordinate ring for Y.

Proof. to be updated

6.107. This immediately says that two affine varieties X and Y are isomorphic if and only if they are isomorphic as k-algebras $A(X) \cong A(Y)$.

6.108. There is an interesting categorical statement here. There is a contravariant functor

$$A(-) : \mathbf{Var}_{\text{affine}} \to \mathbf{Alg}_{\text{k}}^{\text{op}}, \quad Y \mapsto A(Y) \tag{170}$$

where $\mathbf{Alg}_{\text{k}}^{\text{op}}$ is the category of reduced finitely generated $k-$algebras and $\mathbf{Var}_{\text{affine}}$ is the category of affine varieties.

This induces an anti-equivalence of categories $\mathbf{Var}_{\text{affine}} \cong \mathbf{Alg}_{\text{k}}^{\text{op}}$.

REMARK 6.109. For any reduced finitely generated $k-$algebra R, there is an affine variety X such that $R \cong \mathcal{O}(X)$.

6.110. Now we will move to some general discussion on nonsingular varieties. Over $\mathbb{C}$, nonsingular variety has the structure of a complex manifold. Conversely, every smooth complex algebraic variety is nonsingular.

Hence, the concept of nonsingularity is related to do with the smoothness of manifolds, which is why we employ derivatives (and the Jacobian) in their study.

We will also discuss the related GAGA principle.

DEFINITION 6.111 (Nonsingular Affine Variety, Jacobian Criterion). Let X be an affine variety of dimension r, and let $f_1, \cdots, f_t \in A = k[x_1, x_2, \cdots, x_n]$ be the set of generators for ideal of X. We say Y is nonsingular at a point $P \in X$ if the rank of the matrix[11] $||\partial f_t \partial x_j(P)|| = n - r$. If X is nonsingular at each point P, then we say X is nonsingular.

If X is not nonsingular at P, then X is singular at P.

EXAMPLE 6.112. Let $X = V(x^2 - y) \in \mathbf{A}^2$ and a point $P = (a, b)$. To find if P is nonsingular, we find $||\partial f_t \partial x_j(P)||$. Note that $n = 2$ and $r = 1$ since X is an hypersurface defined by one irreducible polynomial. Moreover, the Jacobian at P which we denote by J_P is

$$J_P = [2a \quad -1] \tag{171}$$

for which the rank$(J_P) = 1$, which equals to $n - r = 1$. Hence, X is nonsingular at P.

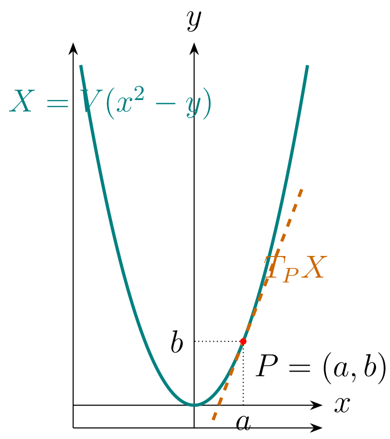

FIGURE 2. Graphic produced using TikZ generated by Claude

[11]The Jacobian matrix. We borrow the notation $||\partial f_t \partial x_j(P)||$ from [25].

6.113. Another definition and theorem (for nonsingular varieties) is in order now.

Recall the definition of local ring (Def. 3.158) and Noetherian ring.

DEFINITION 6.114. Let A be a local Noetherian ring with $\mathfrak{m}$ be the maximal ideal and residue field $k = A/\mathfrak{m}$. We call A is *regular local ring* if[12] $\dim_k \mathfrak{m}/\mathfrak{m}^2 = \dim A$.

THEOREM 6.115. Let X be an affine variety with a point $P \in X$. Then X is *nonsingular* at P if and only if the local ring $\mathcal{O}_p$ is a regular local ring.

We say X is nonsingular if it is nonsingular at every point.

PROOF. Will be added later. Refer to [**25**].

6.116. In fact, the concept of *nonsingularity* is not limited to affine varieties.

DEFINITION 6.117. Let X be any variety with a point $P \in X$. We say X is nonsingular at P if the local ring $\mathcal{O}_P$ is regular local ring. X is nonsingular if it is nonsingular at every point.

X is *singular* if it is not nonsingular.

PROPOSITION 6.118. Let X be any variety. The set X^{Sing} of singular points of X is a proper closed subset.

PROOF. Let us assume that X is affine. We know that any variety can be covered by affine opens.

By the Jacobian criterion, if

$$X = V(f_1, \cdots, f_r) \subseteq \mathbb{A}^n \tag{172}$$

then $P \in X^{Sing} \Leftrightarrow \text{rank } J(P) < n - r$. We know that a matrix has rank $< n - r$ when all of its $(n-r) \times (n-r)$ minors vanish. Since the minors are polynomials, their common zero locus is closed. Hence X^{Sing} is a closed subset.

Now, we have to show that X^{Sing} a proper subset of X. We can show this by contradiction. Let us assume $X^{Sing} = X$. By the Theorem. 6.99, X is birational equivalent to an irreduiclbe hypersurface

$$Y = V(f) \tag{173}$$

for a single non-constant polynomial $f(x_1, \cdots, x_n)$. (The birational varieties are isomorphic on open dense subsets.) The Jacobian criterion states that, for a hypersurface, every point is singular if and only if

$$\frac{\partial f}{\partial x_i}(P) = 0 \tag{174}$$

[12]Generally, it is $\dim_k \mathfrak{m}/\mathfrak{m}^2 \geq \dim A = \text{ht } \mathfrak{m}$

since for hypersurface, we have $r = n - 1$ and so $n - r = 1$ which means that for P to be singular if rank $J_P = 0$.

Now each partial derivative is a polynomial $g_i = \frac{\partial f}{\partial x_i}$, so $g_i \in I(Y)$. Also if Y is an irreducible hypersurface $I(Y) = (f)$, then $g_i \in (f)$ which means that there exists a polynimal h_i such that

$$g_i = h_i f \tag{175}$$

which would mean that $f | g_i$. However this is contradiction, because

$$\deg(g_i) = \deg \frac{\partial f}{\partial x_i} < \deg(f) \tag{176}$$

unless the derivative is identically zero, $\frac{\partial f}{\partial x_i} \equiv 0$. In char $k = 0$, f must be a constant polynomial and hence cannot define a hypersurface. Due to this contradiction there must exist one nonsingular point and we conclude that $X^{Sing} \neq X$.

6.119 (Algebraic and Analytic). The idea, from here, is to discuss the two approaches of studying an algebraic variety; *algebraic* and *analytic*. We will adopt from the GAGA [**31**] of Serre for such purposes. However, we will need to define the definition of sheaves, which was the to be done, as originally planned, in a different chapter of PtoS. Hence, there will a slight overlap in that regard.

Now, a couple of basic definition.

Definition 6.120 (Analytic Set). Let $\Omega \subset \mathbb{C}^n$ be an open set. We define an analytic set A if for every point $p \in \Omega$, there exists an open neighborhood U of p and finitely many holomorphic function $f_1, \ldots, f_r : U \to \mathbb{C}$ such that

$$A \cap U = \{z \in U : f_1(z) = \cdots = f_r(z) = 0\} \tag{177}$$

which means[13] that locally it is the zero locus of the finitely many holomorphic functions.

Example 6.121. The set

$$A = \{z \in \mathbb{C} : z^2 - 1 = 0\} \tag{178}$$

is analytic since $z^2 - 1$ is holomorphic. Another example is the hypersurface

$$A = \{(z, w) \in \mathbb{C}^2 : z^2 + w^3 = 0\}$$

as $z^2 + w^3$ is holomorphic function.

[13]Analytic set is the analogue of *algebraic set* in analytic geometry where polynomial functions are replaced with holomorphic functions.

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